ClipSAT : Linear Algebra Vectors, Matrices, Eigenvalues & Least Squares [ANSWER KEY]

Unit 1.01: Vectors in R^n & Inner Product Spaces

Q1. Find lVertmathbf{u}rVert for mathbf{u}=-9,-12 .

[ ] A: 21
[ ] B: 14
[x] C: 15
[ ] D: 225

Solution: lVertmathbf{u}rVert=sqrt{-9^2+-12^2}=sqrt{225}=15 .

Q2. A 3 times 3 symmetric matrix Q_{k} has eigenvalues lambda_1 = 12 , lambda_2 = 3 , and lambda_3 = 14 . Determine the trace text{tr}Q_{k} .

[ ] A: 26
[ ] B: 32
[x] C: 29
[ ] D: 41

Solution: The trace equals lambda_1 + lambda_2 + lambda_3 = 29 .

Q3. A linear transformation T: mathbb{R}^2 to mathbb{R}^2 is defined by Tmathbf{x} = Amathbf{x} , where A = begin{pmatrix} 2 & 1 1 & 3 end{pmatrix} . If mathbf{v} = begin{pmatrix} 1 2 end{pmatrix} , find Tmathbf{v} .

[x] A: begin{pmatrix} 4 7 end{pmatrix}
[ ] B: begin{pmatrix} 3 7 end{pmatrix}
[ ] C: begin{pmatrix} 4 5 end{pmatrix}
[ ] D: begin{pmatrix} 5 7 end{pmatrix}

Solution: To find Tmathbf{v} , we multiply the matrix A by the vector mathbf{v} . Tmathbf{v} = Amathbf{v} = begin{pmatrix} 2 & 1 1 & 3 end{pmatrix} begin{pmatrix} 1 2 end{pmatrix} . Performing the matrix-vector multiplication: Tmathbf{v} = begin{pmatrix} 2 times 1 + 1 times 2 1 times 1 + 3 times 2 end{pmatrix} = begin{pmatrix} 2 + 2 1 + 6 end{pmatrix} = begin{pmatrix} 4 7 end{pmatrix} . Thus, the correct answer is begin{pmatrix} 4 7 end{pmatrix}.

Q4. Consider the matrix A = begin{pmatrix} 2 & 1 1 & 3 end{pmatrix} . If a point P=x,y is transformed by A to P'=x',y' , such that begin{pmatrix} x' y' end{pmatrix} = A begin{pmatrix} x y end{pmatrix} , what is the image of the point 3,1 under this transformation?

[x] A: 7, 6
[ ] B: 6, 7
[ ] C: 8, 5
[ ] D: 5, 8

Solution: The transformation is defined by the matrix multiplication begin{pmatrix} x' y' end{pmatrix} = A begin{pmatrix} x y end{pmatrix} . We are given the point 3,1 , so begin{pmatrix} x y end{pmatrix} = begin{pmatrix} 3 1 end{pmatrix} and the matrix A = begin{pmatrix} 2 & 1 1 & 3 end{pmatrix} . Performing the matrix multiplication: begin{pmatrix} x' y' end{pmatrix} = begin{pmatrix} 2 & 1 1 & 3 end{pmatrix} begin{pmatrix} 3 1 end{pmatrix} = begin{pmatrix} 2 times 3 + 1 times 1 1 times 3 + 3 times 1 end{pmatrix} = begin{pmatrix} 6 + 1 3 + 3 end{pmatrix} = begin{pmatrix} 7 6 end{pmatrix} Therefore, the image of the point 3,1 is 7,6 .

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Q5. Consider two vectors mathbf{v}_1 = begin{pmatrix} 3 4 end{pmatrix} and mathbf{v}_2 = begin{pmatrix} -4 3 end{pmatrix} . What is the dot product mathbf{v}_1 cdot mathbf{v}_2 ?

[ ] A: 1
[x] B: 0
[ ] C: -7
[ ] D: 25

Solution: text{The dot product of two vectors } mathbf{v}_1 = begin{pmatrix} a b end{pmatrix} text{ and } mathbf{v}_2 = begin{pmatrix} c d end{pmatrix} text{ is given by } mathbf{v}_1 cdot mathbf{v}_2 = ac + bd. text{In this case, } mathbf{v}_1 = begin{pmatrix} 3 4 end{pmatrix} text{ and } mathbf{v}_2 = begin{pmatrix} -4 3 end{pmatrix}. text{Therefore, } mathbf{v}_1 cdot mathbf{v}_2 = 3-4 + 43 = -12 + 12 = 0. text{A dot product of 0 indicates that the vectors are orthogonal.}

Q6. Consider the following system of linear equations represented by the augmented matrix: begin{pmatrix} 1 & 2 & | & 5 2 & -1 & | & 0 end{pmatrix} . Use row reduction to transform the matrix into row-echelon form and determine the unique solution x, y for the system. What is the value of x ?

[x] A: 1
[ ] B: 2
[ ] C: 3
[ ] D: 4

Solution: We are given the augmented matrix: begin{pmatrix} 1 & 2 & | & 5 2 & -1 & | & 0 end{pmatrix} To transform this matrix into row-echelon form, we can perform the following row operation: R_2 leftarrow R_2 - 2R_1 . begin{pmatrix} 1 & 2 & | & 5 2 - 21 & -1 - 22 & | & 0 - 25 end{pmatrix} begin{pmatrix} 1 & 2 & | & 5 0 & -5 & | & -10 end{pmatrix} Now, we can perform another row operation: R_2 leftarrow R_2 / -5 . begin{pmatrix} 1 & 2 & | & 5 0 & 1 & | & 2 end{pmatrix} This matrix is now in row-echelon form. The corresponding system of equations is: x + 2y = 5 y = 2 Substitute the value of y into the first equation: x + 22 = 5 x + 4 = 5 x = 1 Therefore, the unique solution is 1, 2 , and the value of x is 1.

Q7. Consider the vector mathbf{v} = begin{pmatrix} 5 -12 end{pmatrix} . Calculate the magnitude of mathbf{v} , denoted as lVertmathbf{v}rVert .

[x] A: 13
[ ] B: 17
[ ] C: 7
[ ] D: 169

Solution: The magnitude of a vector mathbf{v} = begin{pmatrix} x y end{pmatrix} in mathbb{R}^2 is given by the formula lVertmathbf{v}rVert = sqrt{x^2 + y^2} . For the given vector mathbf{v} = begin{pmatrix} 5 -12 end{pmatrix} , we have x=5 and y=-12 . Therefore, lVertmathbf{v}rVert = sqrt{5^2 + -12^2} = sqrt{25 + 144} = sqrt{169} = 13 .

Q8. Consider the vector mathbf{v} = begin{pmatrix} 3 4 end{pmatrix} . If this vector is plotted on a standard xy -plane, it represents a directed line segment from the origin to the point 3,4 . What is the geometric interpretation of the length of this vector?

[ ] A: The vector's direction in the xy-pla
[ ] B: The angle the vector makes with the
[x] C: The distance of the point 3,4 from t
[ ] D: The sum of the components of the vec

Solution: The length of a vector mathbf{v} = begin{pmatrix} x y end{pmatrix} is given by its magnitude, lVertmathbf{v}rVert = sqrt{x^2 + y^2} . Geometrically, this corresponds to the distance from the origin 0,0 to the point x,y on the Cartesian plane, as described by the Pythagorean theorem. For mathbf{v} = begin{pmatrix} 3 4 end{pmatrix} , the length is sqrt{3^2 + 4^2} = sqrt{9 + 16} = sqrt{25} = 5 . This value represents the distance of the point 3,4 from the origin.